

Problem Set 1

1. Impulse and step representations

The sequence $x[n]$ is $x[-1] = 3$, $x[0] = -2$, $x[2] = 4$, and zero elsewhere.

- a) Express $x[n]$ as a sum of scaled, shifted unit samples.
- b) Let $v[n] = u[n + 1] - u[n - 2]$. List the non-zero samples of $v[n]$ and sketch it.
- c) Express $\delta[n + 2]$ using two shifted unit steps.

2. System-property classification

For each system below, determine whether it is linear, time-invariant, causal, and BIBO stable. Give a one- or two-sentence justification for every "no".

- A. $y[n] = 2x[n - 2]$
- B. $y[n] = x^2[n]$
- C. $y[n] = n x[n]$
- D. $y[n] = x[n] + 0.5 x[n - 1]$

3. Finite convolution and support

Let $x[n] = \{1, -1, 2\}$ for $n = 0, 1, 2$ and $h[n] = \{2, 1\}$ for $n = 0, 1$.

- a) Predict the index range over which $y[n] = x[n] * h[n]$ can be non-zero.
- b) Compute the full convolution by hand.
- c) Verify the results with `np.convolve`.

4. Impulse response, causality, stability, and step response

An LTI system has $h[n] = (0.75)^n u[n]$.

- a) Is the system causal? Explain from $h[n]$.
- b) Is the system BIBO stable? Show the relevant sum.
- c) Find the response to $x[n] = u[n]$. Give a closed-form expression for $n \geq 0$.
- d) What is the limiting output value as $n \rightarrow \infty$?

5. BIBO stability proof for an LTI system (ECEN 5632 problem)

Assume $|x[n]| \leq B_x$ for all n and that $\sum_k |h[k]| < \infty$. Starting from $y[n] = \sum_k x[k]h[n-k]$, prove that $y[n]$ is bounded for all n . State an explicit bound B_y .

Then explain in one sentence why $h[n] = (0.99)^n u[n]$ is stable even though its impulse response lasts forever.